

ANSWER SCHEME

Question 1

Figure 1.1 below shows a variety of cells dividing by mitosis.

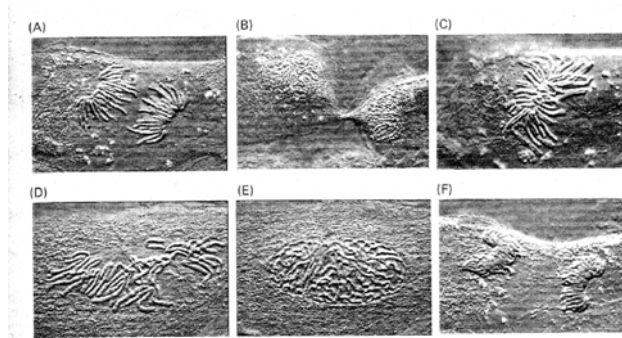


Figure 1.1

- (a) Using the letters assigned to each cell (ie. A, B,), arrange the cells in the correct sequence that they will undergo during division. [1]

- **E, D, C, A, F, B**

- (b) Identify the mitotic stages the following cells are in with reasons. [6]

A

- **Anaphase**
- **Separation of sister chromatids towards opposite poles**

C

- **Metaphase**
- **Chromosomes lined up at the equator of cell**

F

- **Telophase**
- **Chromosomes uncoiled / Cytokinesis beginning**
- **Reject: Chromosomes at opposite poles**

- (c) Discuss the significance of mitotic division to a living organism. [3]

- **Genetic Stability; Produce daughter cells with the same chromosome number and the same genetic information as parent cells [1]**
- **Growth / Regeneration / Replacement; Number of cells in a multicellular environment increases by mitosis [1]**
- **Asexual Reproduction; Asexually reproducing organisms reproduce via mitotic divisions [1]**

[Total: 10 marks]

Question 2

Figure 2.1 illustrates the structure of a typical synapse.

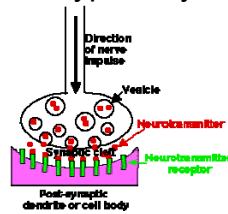


Figure 2.1

- (a) Provide alternative names for the following: [2]

Vesicle – Synaptic Vesicle

Neurotransmitter – Acetylcholine / Noradrenaline

- (b) Describe how an electrical nervous impulse arriving at the pre-synaptic knob could be converted into a chemical form in the synaptic cleft. [3]

- Depolarisation of membrane of synaptic knob / presynaptic membrane
- Opening of calcium channels .
- Influx of calcium ions into synaptic knob
- Fusion of synaptic vesicles with pre-synaptic membrane
- Release of neurotransmitter/acetylcholine into synaptic cleft

Any 2

- (c) Describe how a nervous impulse is generated on the membrane of the post synaptic dendrite. [2]

- Attachment of neurotransmitter to post-synaptic membrane receptor sites
- Depolarisation of post-synaptic membrane
- Opening of sodium channels
- Influx of sodium
- Generation of action potential at post-synaptic membrane

[0.5m each] Any 4

- (d) Chemical X, which has a similar shape to the neurotransmitter, is introduced into the synapse. Explain the effect that it would have on nervous transmission across the synapse. [2]

- Chemical X will bind to post-synaptic membrane receptor sites
- Prevents neurotransmitter from binding
- No action potential is formed at post-synaptic membrane
- Leads to paralysis of muscle / gland

[0.5m each]

- (e) Suggest why synapses are unable to function at low temperatures. [1]

- Enzymes are involved in synaptic transmission
- These enzymes are inactivated at low temperatures.

[0.5m each] [Total: 10 marks]

Question 3

Figure 3.1 is an illustration of a phylogenetic tree.

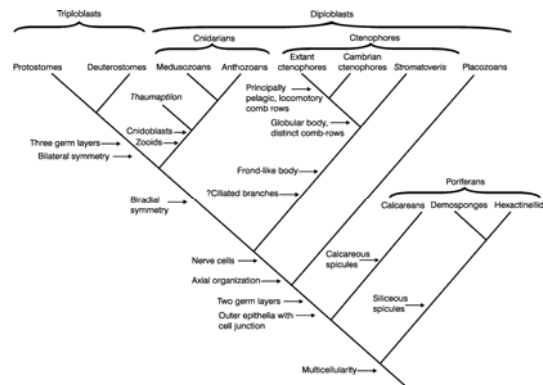


Figure 3.1

(a) Give the general name for such a tree and explain its usefulness. [1]

- **Cladogram**
 - **Allows the study of evolutionary relationships**
- [0.5 m]

(b) Using information from Figure 3.1, suggest and explain if anthozoans are more closely related to medusozoans or protostomes. [2]

- **Anthozoans are more closely related to medusozoans**
- **Branching off point of medusozoans much closer to anthozoans than the branching off point of protostomes**

Figure 3.2 shows the development of finches on the Galapagos Islands.

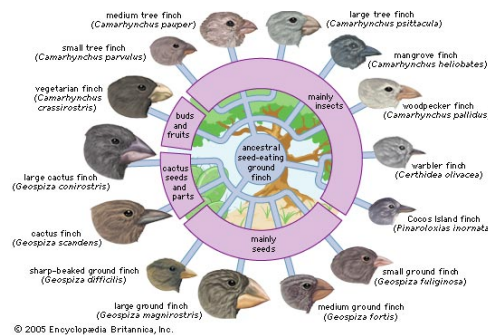


Figure 3.2

- (c) (i) Discuss how the ancestral seed-eating ground finch speciated into a variety of different species. [4]

- **Ancestral finch had variations in beak types**
- **Different beak types allowed for adaptation to different habitats/feeding on different items**
- **Once adapted to the habitat, they did not cross-breed with finches in different habitats**
- **Over time finches in different habitats evolved differently from each other due to the different environment they occupied.**
- **Speciation occurs / Adaptive radiation**

Any 4

- (ii) Compare the type of speciation you just described with sympatric speciation. [2]

- **Both involve sub-populations in the same geographical environment**
- **However, adaptive radiation involves adaptation to separate habitats**

- (d) According to the evolutionary concept of natural selection, recessive disease alleles like those causing sickle cell anaemia and cystic fibrosis should be selected against. This means that the frequency of these alleles in the human population should be low. However, this is not the case. The frequency of these alleles remain relatively high. Suggest a reason for this. [1]

- **Heterozygote advantage / Sometimes the heterozygote is selected for ahead of the two homozygous types; the recessive allele is thus conserved in the heterozygote form**

[Total: 10 marks]

Question 4

- 4 (a)** A diploid plant species is capable of both self- and cross-pollination. Two pure bred varieties **A** and **B** of this plant species were identified. Individuals of variety **A** have heights ranging from 21 to 25 cm. They all have round leaves and are resistant to the herbicide glyphosate. Individuals of variety **B** have heights ranging from 56 to 60 cm. They all have wrinkled leaves and are sensitive to the herbicide glyphosate.

When **A** was crossed with **B**, all F1 progeny had round leaves and was glyphosate resistant. Their height distribution is shown in the table below.

Height / cm	Number of plants
26 – 30	1
31 – 35	10
36 – 40	54
41 – 45	51
46 – 50	8
51 – 55	3

Table 4.1

When the F1 plants were self-pollinated, the F2 progeny had the following phenotypes:

Height / cm	Round leaf & resistant to glyphosate	Wrinkled leaf & resistant to glyphosate	Round leaf & sensitive to glyphosate	Wrinkled leaf & sensitive to glyphosate
21 – 25	15	1	0	5
26 – 30	88	2	2	28
31 – 35	114	4	3	35
36 – 40	152	5	5	48
41 – 45	152	5	6	47
46 – 50	116	4	4	36
51 – 55	86	2	3	27
56 – 60	13	0	0	4
Sub-total	736	23	23	230

Table 4.2

With reference to **Tables 4.1 and 4.2**, answer the following questions

- (i) Are the alleles responsible for wrinkled leaves and glyphosate resistance dominant or recessive? Briefly explain your answer. [2]
- wrinkled leaves - recessive; [1/2]
 - glyphosate resistance- dominant; [1/2]
 - F1 progeny of pure-bred parents A and B were all round leaves and glyphosphate resistance; [1]
- (ii) How many genes and alleles control the glyphosate response? [1]
- One gene [1/2]
 - of 2 different alleles [1/2]
- (iii) With reference to **Tables 4.1 and 4.2**, explain your answer. [2]
- [1]
- From Tables 5.1 and 5.2 → only 2 discrete classes of phenotypes;
OR
 - Either glyphosate resistance or glyphosate sensitive;
- [1]
- Table 5.2 shows 3:1 phenotypic ratio of F2 generation;
OR
 - 3 glyphosate resistant: 1 glyphosate sensitive/ 759:253;
- (iv) Given that the critical value $\chi^2_{(0.05, 3)}$ is 7.815, conduct a chi square analysis to determine if the inheritance of leaf shape and glyphosate response follows Mendel's law of independent assortment. Refer to Annex A for the formula. [5]

Let the H_0 of the cross between the round leaf & glyphosate resistant and wrinkled leaf & glyphosate sensitive [1/2] be in the diybrid inheritance (independently assorted) in the

ratio 9:3:3:1 [1/2].

OR

Let H_0 be the inheritance of leaf shape and glyphosate response [1/2] that follows the dihybrid inheritance (Mendel's law of independent assortment) of ratio of 9:3:3:1 [1/2].

Class	H_0	Observed No. (O)	Expected No. (E)	(O – E)	$\frac{(O - E)^2}{E}$
Round leaf/ glyphosate resistant	9/16	736	569.25	166.75	48.85
Wrinkled leaf/ glyphosate resistant	3/16	23	189.75	-166.75	146.54
Round leaf/ glyphosate sensitive	3/16	23	189.75	-166.75	146.54
Wrinkled leaf / glyphosate sensitive	1/16	230	63.25	166.75	439.61

Correct table [1]

One mistake [-1/2]

Two or more mistakes [no marks]

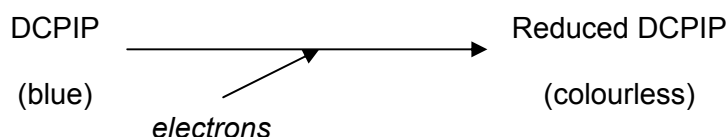
$$\chi^2 \text{ cal} = 781.54 \quad [1/2]$$

$$\chi^2 \text{ cal} = 781.54 > 7.815 \quad [1/2]$$

Marking Points	Marks
Since $\chi^2 \text{ cal} = 781.54 > 7.815$, the calculated χ^2 value is <u>higher</u> than the critical value	½
the difference between the observed and expected results is <u>statistically significant</u>	½
H_0 is rejected.	½
The phenotypic ratio of leaf shape and glyphosate response does not fit Mendel's law of independent assortment ratio of 9:3:3:1	½

Question 5

2,6-dichlorophenolindophenol (DCPIP) is a blue dye that can be converted to colourless reduced DCPIP by gaining electrons. This is summarized below.



A chloroplast suspension was made by grinding fresh leaves in buffer solution and centrifuging the mixture. Tubes were prepared and treated in different ways. The colour of the tube contents was recorded at the start and after 15 minutes. This information is summarized in the table.

Tube	Contents	Treatment	Colour	
			At start	After 15 minutes
A	2cm ³ chloroplast suspension 6cm ³ DCPIP	Tube kept in bright light	Blue/ green	Green
B	2cm ³ chloroplast suspension 6cm ³ DCPIP	Tube kept in dark	Blue/ green	Blue/ green
C	2cm ³ buffer solution 6cm ³ DCPIP	Tube kept in bright light	Blue	Blue

- (a) (i) Tube C was included as a control. Explain why this control was necessary in the investigation. [1]
To serve as a reference point to show that chloroplast was responsible for the colour change [1]

- (ii) Explain the colour of tube A after 15 minutes. [2]

-Light is absorbed by accessory pigments in light harvesting complex and passed down to reaction centre.[1/2]
-Chlorophyll a in reaction centre release an electron.[1/2]
-Electron reduce blue DCPIP to colourless DCPIP.
-Green chloroplast visible. [1/2]

- (b) Describe how reduced NADP is involved in the light-independent reactions of photosynthesis. [2]

-Acts as a source of electron. [1]

-Reduces glycerate-3-phosphate to triose phosphate.[1]

(note: Short form not accepted)

- (c) Apart from the difference in the presence or absence of oxygen, list a difference between aerobic and anaerobic respiration. [1]

-Any one below:

-In aerobic respiration, 38 ATP is obtained from 1 molecule of glucose whereas in anaerobic respiration 2 ATP is obtained. [1]

-End products of aerobic respiration: CO₂, H₂O, ATP

End products of anaerobic respiration: ATP, lactate (mammals), ethanol (yeast) [1]

- (d) During glycolysis, NAD is reduced. Explain what happens to this reduced NAD when the cell is respiring anaerobically. [2]

-NADH accumulates in cytoplasm because oxidative phosphorylation ceases and NADH cannot be oxidised to NAD⁺. [1]

-To be reoxidised, NADH must give H atom to alternate H acceptor eg pyruvate (mammal) or ethanol (yeast). [1]

- (e) Explain why, in the absence of oxygen, the electron transfer chain in the mitochondria does not operate. [2]

-O² is the final electron acceptor of ETC. [1/2]

-If O² not available, electron carrier cannot be re-oxidised/remain in reduced state.[1/2]

-Electron carrier cannot pass down electron to the next carrier below them.[1/2]

-Reduced FAD and NAD cannot be oxidised. [1/2]

Question 6

The following figure 6.1 is a simplified diagrammatic representation of a cell signalling pathway.

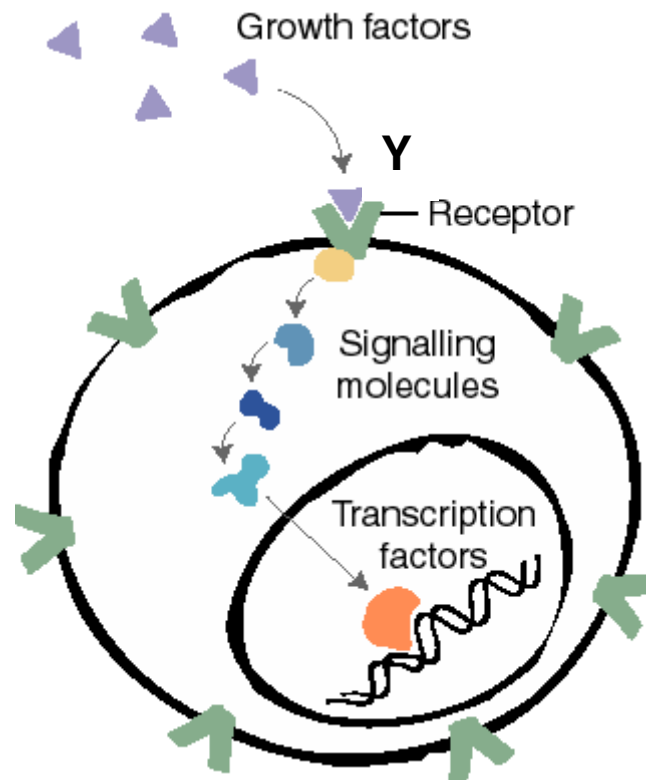


Figure 6.1

(a) With the use of a specific example, discuss the role of structure Y. [3]

- **Binding site for growth factor**
- **A ligand-receptor interaction is created**
- **Binding changes conformation of receptor**
- **Causes it to activate another cellular molecule**
- **This molecule will serve as a signaling molecule**
- **Example of G-protein linked receptor**

[0.5 m each]

(b) Suggest why transcription factors are the normal end products of such a pathway. [1]

- **Because the response would require the production of proteins / transcription factors will initiate transcription resulting in mRNA for translation into proteins**

Figure 6.2 shows a simplified representation of the downstream cellular effects of the dimerization and activation of the VEGF/VEGFR-2 complex in an endothelial cell.

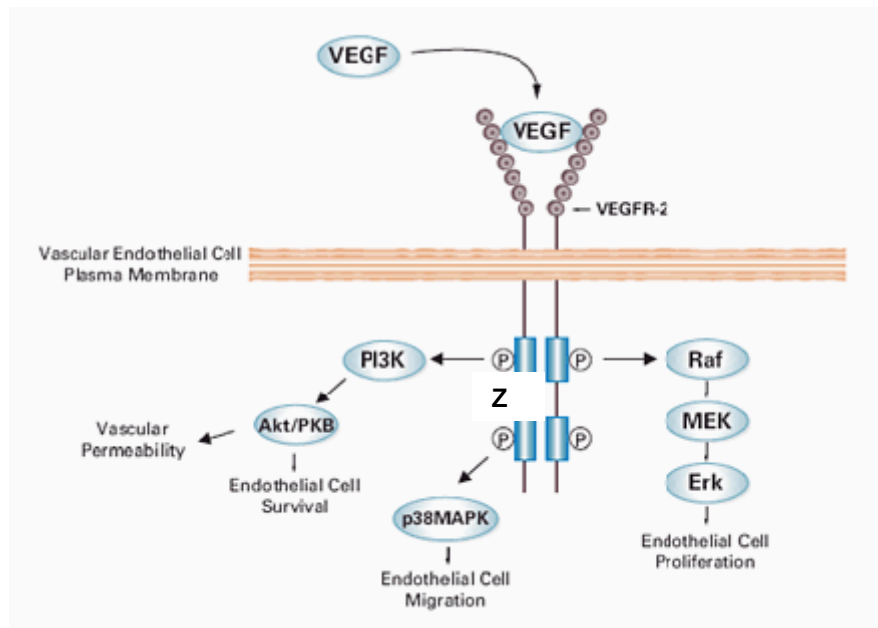


Figure 6.2

(c) Label clearly on the diagram the three main stages of cell signalling. [1]

- Labels required for 'Reception', 'Signal Transduction' and 'Cellular Response'

(d) Suggest a role for the phosphorylated molecules labelled Z. [2]

- Creates a phosphorylation cascade
- To amplify the signal

(e) Using information from Figure 6.2 explain why this signal transduction pathway may lead to cancerous expression of the endothelial cell. [3]

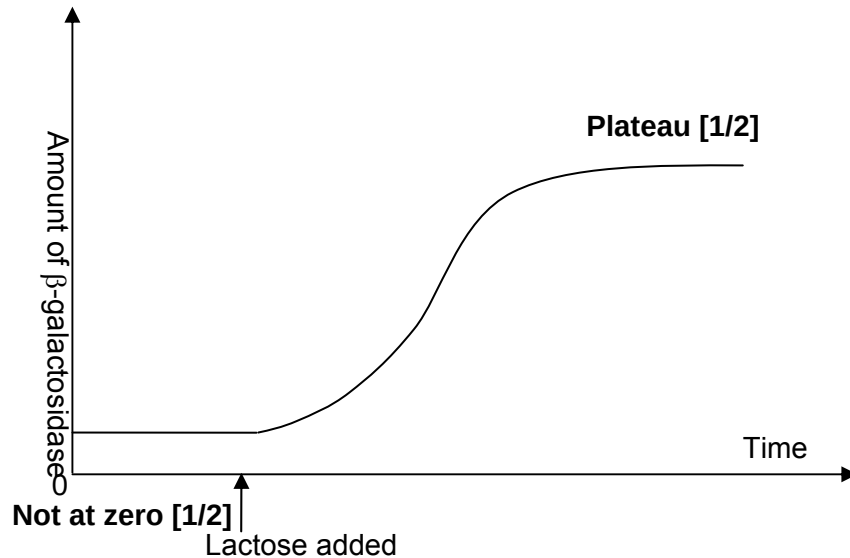
- Because cell responses include:
 - o Cell proliferation, may lead to tumour formation
 - o Cell migration, may lead to metastasis
 - o Cell survival, may lead to immortality
 - o Increased vascular permeability, may lead to angiogenesis

Any 4

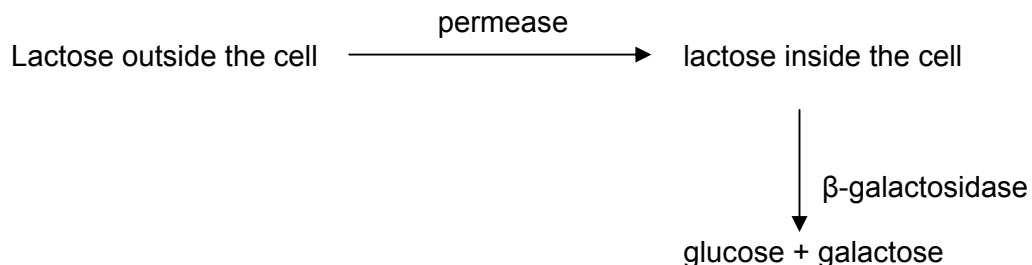
[Total: 10 marks]

Question 7

- (a) A study of the bacterium's chromosome showed that it possessed a *lac* operon to allow the bacterium to metabolise lactose.
- (i) On the axes below, draw a graph showing how the amount of β -galactosidase in the bacterium is affected by the addition of lactose. [1]



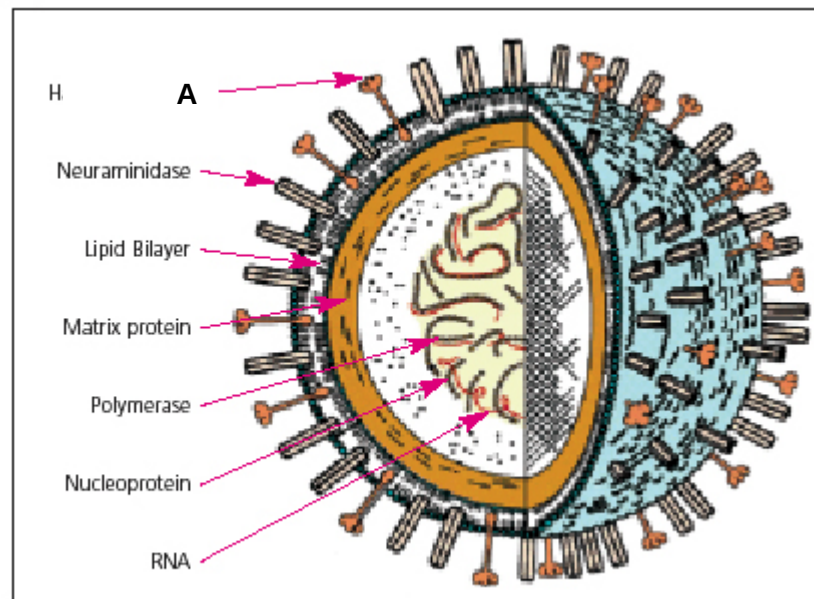
- (ii) Describe how the following pathway is turned on in *E.coli*. [4]



[1 mark each, max 4]

- permease coded by *lacY*, galactosidase coded by *lacZ* and transacetylase coded by *lacA*
- Regulator gene (*lacI*) codes for a repressor that binds to the operator, blocking the attachment of RNA polymerase
- In the presence of lactose, allolactose binds to the repressor protein and alters its conformation/configuration, hence, inability of repressor binding to the operator
- RNA polymerase is not blocked and thus transcription of *lac* mRNA by RNA polymerase occurs
- When glucose is scarce, cAMP binds to inactive cAMP receptor protein (CRP)/catabolite activator protein (CAP)
- Active CRP binds to CRP binding site on the DNA which enhances the binding of RNA polymerase to the promoter, increasing transcription

(b) Fig. 7.1 shows the main structural features of an influenza virion.



Source: Chotani RA (2006) *The impact of pandemic influenza on public health*.¹⁶

Fig. 7.1

(i) Identify molecule **A** and describe its role in infection. [2]

[1 mark each, max 2]

- **Haemagglutinin (preferred spelling)**
- **Haemagglutinin on the virus binds to the sialic acid receptors of the host cell.**

(ii) Describe the role of neuraminidase in the life cycle of the influenza virus. [1]

[1/2 mark]

- **During release, neuraminidase remove sialic acid from the surface of the new viral particle**
- **To prevent haemagglutinin molecule on one particle from attaching to the sialic acid on others that result in clumping**

(c) Table 7.1 shows an excerpt of the confirmed human cases of avian influenza reported by the World Health Organisation (WHO).

Table 7.1: Cumulative Number of Confirmed Human Cases of Avian Influenza A/(H5N1) reported to WHO 19 June 2008

Country	2003		2004		2005		2006		2007		2008		cases
	cases	deaths	cases	deaths	cases	deaths	cases	deaths	cases	deaths	cases	deaths	
Azerbaijan	0	0	0	0	0	0	8	5	0	0	0	0	8
Bangladesh	0	0	0	0	0	0	0	0	0	0	1	0	1
Cambodia	0	0	0	0	4	4	2	2	1	1	0	0	7
China	1	1	0	0	8	5	13	8	5	3	3	3	30
Djibouti	0	0	0	0	0	0	1	0	0	0	0	0	1
Egypt	0	0	0	0	0	0	18	10	25	9	7	3	50
Indonesia	0	0	0	0	20	13	55	45	42	37	18	15	135
Iraq	0	0	0	0	0	0	3	2	0	0	0	0	3

(i) Avian flu is typically described as “influenza caused by viruses adapted to birds”.
Explain

the occurrence of human cases of avian influenza.

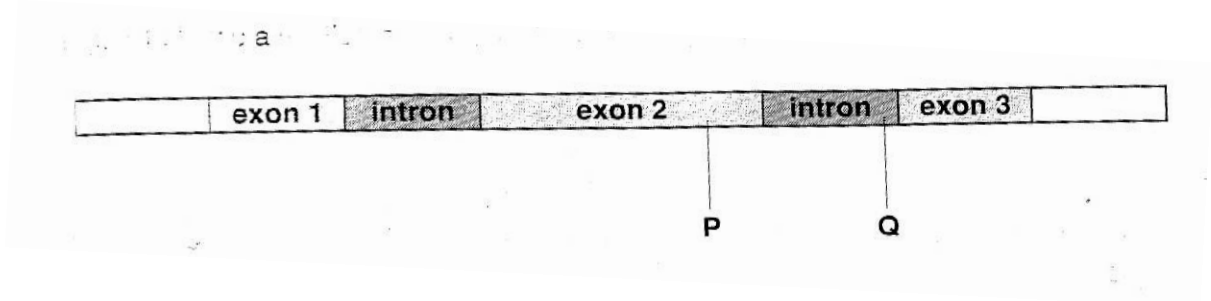
[2]

[1/2 mark each, max 2]

- **Caused by antigenic shift**
- **Process by which two different strains of influenza combine to form a new subtype having a mixture of the surface antigens of the two original strains**
- **Antigenic shift may result when a human influenza virus acquires a new genome segment from an influenza virus capable of infecting other animals such as ducks**
- **When two different strains of influenza infect the same cell simultaneously, their protein capsids and lipid envelopes are removed, exposing their RNA, which is then transcribed to mRNA. The host cell then forms new viruses that combine antigens**
- **This new genome segment causes a major change in the hemagglutinin antigen.**
- **Antibodies made against the original influenza virus can no longer bind to the new strain of virus or stick the virus to phagocytes.**

Question 8

Figure 3.1 shows a section of DNA, containing one gene.

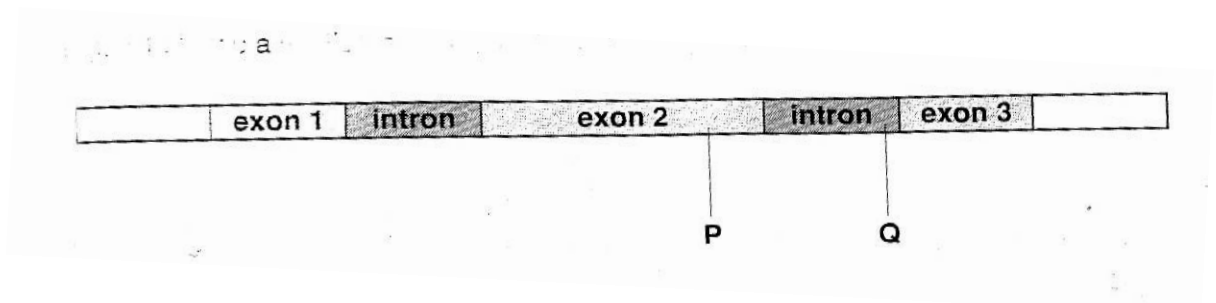


- (a) (i) State one way in which the structure of this gene indicates that it is from a eukaryote. [1]

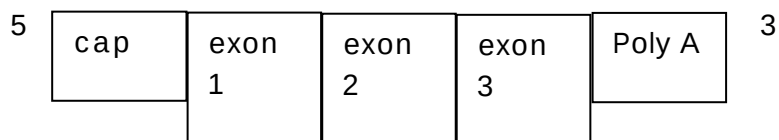
-Presence of introns. [1]

- (ii) Using the same diagrammatic format as Fig 3.1, show what the pre-mRNA and mature mRNA will be for this gene.

Pre-mRNA [1]



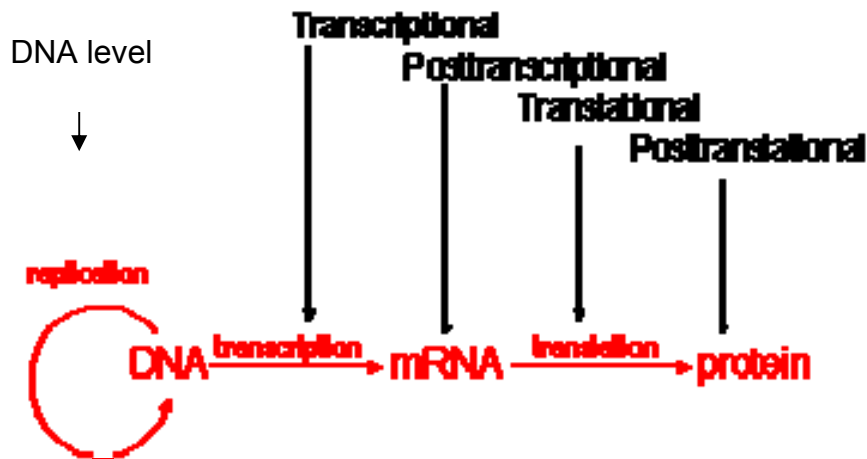
Mature mRNA [2]



(iii) Name and describe the process that has occurred when a pre-mRNA is converted to a mature mRNA. [3]

- **Splicing[1]**
- **Spliceosome made up of both RNA and protein molecules which recognize the splice sites**
- **will cut at the splice site at the end of each intron and then immediately joins together the two exons flanking the intron. [1]**

Control of gene expression can occur at any step: any step: at the DNA level, transcriptional level, post-transcriptional, translational level and post-translational level.



- State two ways to control the expression of genes at the DNA level and explain how these modifications work to control gene expression. [4]

-DNA methylation [1]

- attachment of methyl groups (-CH₃) to DNA bases prevent attachment of RNA polymerase , transcription cannot occur. [1]

-histone acetylation [1]

-When the histones of a nucleosome are acetylated, they change shape so that they grip the DNA less tightly, transcription cannot occur. [1]

Section B

Question 9

(a) Briefly explain why membranes are described as fluid mosaic in structure. [6]

- Made up of a lipid bilayer
- Fluid means that membrane is in constant motion
- Has an oil-like consistency
- Mosaic means that proteins are embedded within the membrane
- In a random manner
- Proteins can be extrinsic or intrinsic

(b) Describe how the membrane affects the entry and exit of substances in and out of a cell. [6]

- Membrane is selectively permeable, due to the lipid bilayer and membrane proteins
- Hydrophobic / non-polar substances are able to dissolve in the lipid bilayer, and pass through the membrane down a concentration gradient
- Small molecules such as water and gases are able to pass through the membrane without dissolving
- Ions, hydrophilic/polar molecules and large molecules are unable to pass through the fatty acid tails of the bilayer
- Presence of trans-membrane proteins which selectively allow certain polar molecules and ions to pass through
- May occur down or against a concentration gradient (active transport)

(c) Using appropriate examples, describe how substances can be brought across a membrane by facilitated diffusion, active transport and endo/exo-cytosis. [8]

Facilitated Diffusion

- Involves carrier or channel proteins
- Allows polar molecules and ions to pass through membrane
- Occurs down a concentration gradient / Does not involve energy
- Substance is brought in by a change in conformation of a carrier protein
- Substance passes through a hydrophilic channel for channel proteins

Max 3

Active Transport

- Involves carrier proteins
- Allows polar molecules and ions to pass through membrane
- Occurs against a concentration gradient
- Requires energy in the form of ATP
- Substance is brought in by a change in conformation of a carrier protein

Max 3

Endo/Exo-cytosis

- Also known as 'Bulk Transport'
- For endocytosis, membrane invaginates to form vesicles containing liquids (endocytosis) or solids (endocytosis)
- For exocytosis, vesicles fuse with the membrane to release their contents outside the cell (release of liquids = exocytosis and release of solids = exocytosis)

Max 2 [Total: Max 8]

Question 10

(a) Describe the structure of DNA & its arrangement in a chromosome. [8]

[1/2 mark each, max 8]

- Double helix;
- 2 polynucleotides strands
- Chains run in **opposite** directions: 3' end of one chain lies opposite the 5' end of the other
→ **anti-parallel**
- Each nucleotide consists – deoxyribose sugar, phosphoric acid and nitrogenous base;
- 4 bases - Adenine, Thymine, Cytosine and Guanine;
- Bases project at **right angle** inward;
- 0.34nm between adjacent base pairs;
- Hydrogen bonds between complementary base pairs
- 2 hydrogen bonds between adenine and thymine, 3 hydrogen bonds between cytosine and guanine;
- Each base pair = one purine + one pyrimidine
- Sugar-phosphate backbone with phosphodiester links;
- (width between the two sugar-phosphate backbone 0.2nm);
- 10 bases pairs in one complete turn/ 3.4nm per turn;
- 2 turns of DNA on histone octamer → nucleosome
- formation of 'beads on a string'
- Strings of nucleosomes coiled to form a 30nm chromatin fiber (solenoid);
- Loop domains of the 30nm fiber/ 30nm fiber forms loops;
- Attached to a scaffold of non-histone proteins;
- Progressively looping and coiling to form metaphase chromosome;

- (b) Distinguish between DNA replication and transcription, indicating the important features of each process.

[6]

1 mark for each feature and the corresponding differences [max 6 marks]

Features	DNA replication	Transcription
(Main) Enzyme involved	DNA polymerase - binds to the parental DNA molecule @ the origin of replication	RNA polymerase - binds to the promoter. a region near the beginning of the gene to be transcribed
Raw Material	Deoxyribonucleotides	Ribonucleotides
Template	Both strands of DNA molecule act as a template Whole DNA molecule is being replicated	One strand of DNA molecule acts the template The anti-sense / non-coding / template strand of DNA molecule is being transcribed; the other strand is not transcribed
Base Pairing	Adenine with thymine and vice versa Cytosine with guanine and vice versa	Adenine with uracil and vice versa Cytosine with guanine and vice versa
Presence of proofreading property on enzyme involved	DNA polymerase proofreads the newly synthesized daughter strand, ensuring precise complementary base pairing	NIL
Products	2 DNA molecules. Each DNA molecule comprises of 1 parental strand and 1 complementary daughter strand Products remain in the nucleus	1 RNA molecule Products leave the nucleus
When the process occur	Prior to nuclear division (mitosis and meiosis)	Prior to translation
Purpose	Double the amount of DNA so that: after mitosis - the 2 daughter cells will have the same amt of DNA (2n) as parental cell after meiosis - the 4 daughter cells each will have half (n) the amt of DNA	Protein synthesis

- (c) Explain, with an example, how a change in the sequence of the DNA nucleotide may affect the amino acid sequence in a protein and hence the phenotype of the organism.

[6]

Explain using either Sickle-cell anaemia OR phenylketonuria OR SCID OR cystic fibrosis

Sickle-cell anaemia [1/2 mark each, max 6]

- Sickle-cell anaemia is a result of a gene or point mutation;
- In sufferers of sickle cell anaemia, haemoglobin S (Hb S) is produced instead of haemoglobin A;
- This is due to a single base substitution;
- thymine is being replaced by adenine in the gene coding for the β -globin chain;
- In the mRNA produced, one of the codons is changed from GAA (codon coding for amino acid glutamic acid) to GUA (codon coding for amino acid valine);
- The resultant β -globin chain has a valine residue instead of a glutamic acid residue;
- Valine is hydrophobic whereas glutamic acid is hydrophilic;
- This substitution thus decreases the solubility of deoxygenated Hb S;
- When the level of oxygen in blood is low, the Hb S molecules precipitate out of solution to form rigid fibres;
- Rigid fibres distort and damage the red blood cells;
- This causes red blood cells to change from a circular biconcave shape to a sickle shape;
- Sickle-cell red blood cells are less efficient in delivery of oxygen to tissues;
- May block fine capillaries due to the sickle shape;
- prevents the normal blood flow;
- sickle red blood cells are more fragile than normal red blood cells;
- sickle red blood cells haemolyse readily and this results in anaemia.

Phenylketonuria [1/2 mark each, max 6]

- Phenylketonuria (PKU) is an example of an inborn error of amino acid metabolism.
- Phenylalanine is one of ten essential amino acids that cannot be synthesised by the body and must be consumed in the diet.
- Classical PKU is a homozygous recessive disorder
- caused by mutations in both alleles of the gene for phenylalanine hydroxylase, which is found on chromosome 12.
- sufferers of PKU (or phenylketonurics) lack the enzyme phenylalanine hydroxylase.
- They are unable to convert phenylalanine to tyrosine.
- Tyrosine is essential for the synthesis of the black pigment melanin.
- Hence, the phenylketonurics will have lighter skin and hair colour than normal individuals
- and also suffer from mental retardation.
- Since phenylketonurics cannot convert phenylalanine to tyrosine, there is an accumulation of phenylalanine in all body fluids.
- Phenylalanine is converted to phenylpyruvate, which is then converted to other compounds.
- Mental retardation is probably due to the derivatives of phenylpyruvate.

SCID [1 mark each, max 6]

- S1 more males affected (than females);
- S2 recessive;
- S3 (most common form is) sex-linked (X chromosome);
- S4 mutation of IL2RG (interleukin-2 receptor gamma) gene (for gamma chain on lymphocyte receptor);
- S5 A mutation in JAK3, located on chromosome 19;
- S6 ADA (adenosine deaminase) deficiency);
- S7 mutation of gene on chromosome 20;
- S8 prevents formation of, lymphocytes/ T cells/ B cells;
- S9 toxic metabolites not broken down so lymphocytes killed;
- S10 so susceptible to opportunistic infections;
- S11 children usually die (if not treated);
- S12 pneumonia/ PCP/ meningitis/ chickenpox/ measles/ other valid;

[must link S8 to S10 or S9 to S10]

Cystic fibrosis [1 mark each, max 6]

- C1 autosomal/ chromosome 7;
- C2 recessive;
- C3 mutation of, **cystic fibrosis transmembrane conductance regulator gene**;
- C4 deletion of, 3 base (pairs)/ codon;
- C5 CFTR/ CFTCR, (protein) has lost phenylalanine;
- C6 (channels) no longer able to transport chloride ions; [ignore direction of Cl⁻ ions]
- C7 causing thick/ sticky mucus;
- C8 resulting in stated problem in, lungs/ pancreas/ reproductive system;
- C9 reduces life expectancy;

[must link S7 to S8]